```python
# 1. IMPORT LIBRARIES
import pandas as pd
import numpy as np
from google.colab import files
files.upload()
from google.colab import files
files.download("featured_housing.csv")


# 2. LOAD DATASET
df = pd.read_csv("/content/housing.csv")

print("Original Data:")
print(df.head())

# 3. HANDLE MISSING VALUES
# Fill numeric with median
df.fillna(df.median(numeric_only=True), inplace=True)

# Fill categorical with mode
for col in df.select_dtypes(include='object').columns:
    df[col].fillna(df[col].mode()[0], inplace=True)


# 4. FEATURE ENGINEERING (NEW FEATURES)


# Avoid division errors
df["rooms_per_household"] = df["total_rooms"] / df["households"]
df["bedrooms_per_room"] = df["total_bedrooms"] / df["total_rooms"]
df["population_per_household"] = df["population"] / df["households"]


# 5. HANDLE INFINITE VALUES
df.replace([np.inf, -np.inf], np.nan, inplace=True)
df.fillna(df.median(numeric_only=True), inplace=True)


# 6. LOG TRANSFORMATION (REDUCE SKEWNESS)
df["log_median_income"] = np.log1p(df["median_income"])
df["log_total_rooms"] = np.log1p(df["total_rooms"])
df["log_population"] = np.log1p(df["population"])


# 7. BINNING (CATEGORICAL CREATION)
df["income_category"] = pd.cut(
    df["median_income"],
    bins=[0, 2, 4, 6, np.inf],
    labels=["Low", "Medium", "High", "Very High"]
)


# 8. ENCODE CATEGORICAL VARIABLES
df = pd.get_dummies(df, drop_first=True)


# 9. FINAL CHECK
print("\nColumns after Feature Engineering:")
print(df.columns)

print("\nMissing Values:", df.isnull().sum().sum())

print("\nSample Data:")
print(df.head())


# 10. SAVE FINAL DATASET
df.to_csv("/content/drive/MyDrive/Machine Learning - Project/featured_housing.csv", index=False)

print("\nFEATURE ENGINEERING COMPLETED SUCCESSFULLY")
```

Choose files housing.csv
housing.csv(text/csv) - 1423529 bytes, last modified: 17/04/2026 - 100% done

```
Saving housing.csv to housing (4).csv
Original Data:
   longitude  latitude  housing_median_age  total_rooms  total_bedrooms  \
0    -122.23     37.88                41.0        880.0           129.0
1    -122.22     37.86                21.0       7099.0          1106.0
2    -122.24     37.85                52.0       1467.0           190.0
3    -122.25     37.85                52.0       1274.0           235.0
4    -122.25     37.85                52.0       1627.0           280.0

   population  households  median_income  median_house_value ocean_proximity
0       322.0       126.0         8.3252            452600.0        NEAR BAY
1      2401.0      1138.0         8.3014            358500.0        NEAR BAY
2       496.0       177.0         7.2574            352100.0        NEAR BAY
3       558.0       219.0         5.6431            341300.0        NEAR BAY
4       565.0       259.0         3.8462            342200.0        NEAR BAY

Columns after Feature Engineering:
Index(['longitude', 'latitude', 'housing_median_age', 'total_rooms',
       'total_bedrooms', 'population', 'households', 'median_income',
       'median_house_value', 'rooms_per_household', 'bedrooms_per_room',
       'population_per_household', 'log_median_income', 'log_total_rooms',
       'log_population', 'ocean_proximity_INLAND', 'ocean_proximity_ISLAND',
       'ocean_proximity_NEAR BAY', 'ocean_proximity_NEAR OCEAN',
       'income_category_Medium', 'income_category_High',
       'income_category_Very High'],
      dtype='object')

Missing Values: 0

Sample Data:
   longitude  latitude  housing_median_age  total_rooms  total_bedrooms  \
0    -122.23     37.88                41.0        880.0           129.0
1    -122.22     37.86                21.0       7099.0          1106.0
2    -122.24     37.85                52.0       1467.0           190.0
3    -122.25     37.85                52.0       1274.0           235.0
4    -122.25     37.85                52.0       1627.0           280.0

   population  households  median_income  median_house_value  \
0       322.0       126.0         8.3252            452600.0
1      2401.0      1138.0         8.3014            358500.0
2       496.0       177.0         7.2574            352100.0
3       558.0       219.0         5.6431            341300.0
4       565.0       259.0         3.8462            342200.0

   rooms_per_household  ...  log_median_income  log_total_rooms  \
0             6.984127  ...           2.232720         6.781058
1             6.238137  ...           2.230165         8.867850
2             8.288136  ...           2.111110         7.291656
3             5.817352  ...           1.893579         7.150701
4             6.281853  ...           1.578195         7.395108

   log_population  ocean_proximity_INLAND  ocean_proximity_ISLAND  \
0        5.777652                   False                   False
1        7.784057                   False                   False
2        6.208590                   False                   False
3        6.326149                   False                   False
4        6.338594                   False                   False

   ocean_proximity_NEAR BAY  ocean_proximity_NEAR OCEAN  \
0                      True                       False
1                      True                       False
2                      True                       False
3                      True                       False
4                      True                       False

   income_category_Medium  income_category_High  income_category_Very High
0                   False                 False                       True
1                   False                 False                       True
2                   False                 False                       True
3                   False                  True                      False
4                    True                 False                      False

[5 rows x 22 columns]
/tmp/ipykernel_6172/2243136647.py:22: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col]


  df[col].fillna(df[col].mode()[0], inplace=True)

FEATURE ENGINEERING COMPLETED SUCCESSFULLY
```

Start coding or generate with AI.